

Ch. 15 — P01 (D) Designing a Digital Twin for a Manufacturing Cell

Problem Bank | Chapter 15 | Type: D | Difficulty:

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Background

A **digital twin** is a continuously updated simulation model that mirrors a physical system in near-real time. Unlike a traditional simulation study (run once, report results), a digital twin ingests live sensor data, maintains current state, and can run ahead-of-time (“lookahead”) to support operational decisions.

Scenario

A precision machining cell consists of:

- 3 CNC machines (M1, M2, M3) arranged in series: all parts visit $M1 \rightarrow M2 \rightarrow M3$.
- Each machine has a probabilistic failure mode: $MTTF \approx 40$ hr, $MTTR \approx 2$ hr (both exponential).
- Processing times: $M1 \sim \text{LogNormal}(\mu = 2.5, \sigma = 0.3)$ min; $M2 \sim \text{Gamma}(3, 1.2)$ min; $M3 \sim \text{Exponential}(4.0)$ min.
- Arrival rate: 8 parts/hr (Poisson).
- Sensor data available: part completions at M1, M2, M3 (timestamped); machine state (idle/busy/down) at 1 Hz.

Design Tasks

- Architecture diagram:** Draw (sketch or describe) the data flow from physical sensors $\rightarrow$ state estimator $\rightarrow$ SimPy digital twin $\rightarrow$ decision dashboard. Identify: the data ingestion pipeline, the state synchronization module, the lookahead runner, and the output layer.
- State synchronization:** When the physical cell sends a “M2 part-complete” event at timestamp t_{sensor} , how should the digital twin update? List the SimPy objects that must be modified (queue lengths, resource states, environment clock) and describe a protocol for handling events that arrive out of order (e.g., a message delayed 500 ms).
- Lookahead policy:** The twin runs 15-minute lookahead simulations on demand to answer: “If M1 fails

now, what is the expected throughput loss in the next 15 minutes?” Design the lookahead procedure:

- How do you snapshot the current twin state?
 - How do you inject the hypothetical M1 failure?
 - How many lookahead replications are needed for a 95% CI with half-width ≤ 1 part? (Use a pilot estimate: std dev of 15-min throughput ≈ 2.5 parts.)
- (d) **Drift detection:** Over time, the real system may drift from the model (e.g., tools wear and increase processing time). Propose a statistical test that the digital twin could run continuously to detect when model parameters have drifted significantly from observed data. Specify: the null hypothesis, the test statistic, the detection threshold, and what action to take upon detection.
- (e) **Limitations:** State three inherent limitations of the digital twin approach that cannot be eliminated by better data or more compute. For each, describe its impact on decision quality.

Deliverables

Write your answer as a structured design document with sections (a)–(e). Part (c) must include a numerical calculation for the required number of replications. Part (d) must name a specific statistical procedure (e.g., CUSUM, Kolmogorov-Smirnov, Welch’s t -test) and justify the choice.